# Neonatal Point-of-Care Ultrasound Course

Preliminary Programme — 31 May to 3 June 2027

*Preliminary — subject to change*

## Course overview

The course runs over four days for 60 participants and covers four topics: cardiac, lung, GI/abdominal and vascular access ultrasound. Days 1–3 are the lecture programme; Day 4 is entirely hands-on.

| Day | Date | Main focus |
|---|---|---|
| Day 1 | Monday 31 May | Cardiology and haemodynamics, and GI ultrasound (incl. NEC) |
| Day 2 | Tuesday 1 June | Lung ultrasound, vascular access, GI ultrasound and ultrasound-guided lumbar puncture; PDA assessment and management (online) |
| Day 3 | Wednesday 2 June | Vascular access, GI ultrasound, the crashing baby protocol and neuro (sinus venous thrombosis), haemodynamics in asphyxia; elective workshops |
| Day 4 | Thursday 3 June | Hands-on scanning and small-group discussions |

## Day 1 — Monday 31 May 2027

### Cardiology, Haemodynamics and GI Ultrasound

| Time | Area | Topic | Duration |
|---|---|---|---|
| *08:00–08:30* | — | *Registration and welcome coffee* | *30 min* |
| *08:30–08:45* | — | *Welcome, course aims and structure* | *15 min* |
| 08:45–09:25 | General | Ultrasound Physics and Knobology: Optimising the Neonatal Image | 40 min |
| *09:25–09:45* | — | *Coffee break* | *20 min* |
| 09:45–10:30 | Cardiac | Neonatal Echocardiography I: Standard Views and Normal Anatomy | 45 min |
| 10:30–11:10 | GI | Introduction to Abdominal POCUS: Systematic Approach and Normal Anatomy | 40 min |
| 11:10–11:50 | Cardiac | Neonatal Echocardiography II: Assessing Cardiac Function and Filling | 40 min |
| 11:50–12:35 | GI | Bowel Ultrasound and Necrotising Enterocolitis (NEC) — Prof. Alan Daneman, SickKids, Toronto | 45 min |
| *12:35–13:35* | — | *Lunch break* | *60 min* |
| 13:35–14:05 | Cardiac | Tamponade and Pericardial Effusion | 30 min |
| 14:05–14:50 | GI | Malrotation, Volvulus and Other Surgical Abdominal Emergencies | 45 min |
| *14:50–15:10* | — | *Coffee break* | *20 min* |
| 15:10–15:55 | Cardiac | Pulmonary Hypertension and Right Ventricular Function | 45 min |
| *15:55–16:30* | — | *Questions, discussion and close of Day 1* | *35 min* |

# Day 2 — Tuesday 1 June 2027

## Lung Ultrasound, Vascular Access, GI and Lumbar Puncture, with the PDA

| Time | Area | Topic | Duration |
|---|---|---|---|
| *08:00–08:20* | — | *Welcome to Day 2 and recap of Day 1* | *20 min* |
| 08:20–09:10 | Lung | ABC of Lung Ultrasound: Patterns and Findings | 50 min |
| 09:10–09:50 | Vascular access | Introduction to Ultrasound-Guided Vascular Access: Vascular Anatomy and Vessel Assessment | 40 min |
| *09:50–10:10* | — | *Coffee break* | *20 min* |
| 10:10–10:55 | Vascular access | Ultrasound-Guided Insertion Techniques and Needle Guidance | 45 min |
| 10:55–11:40 | GI | Renal and Urinary Tract Ultrasound | 45 min |
| 11:40–12:20 | Lung | LUS in the Neonatal Unit | 40 min |
| *12:20–13:20* | — | *Lunch break* | *60 min* |
| 13:20–14:00 | Procedures | Ultrasound-Guided Lumbar Puncture: Spinal Anatomy, Landmarks and Needle Guidance | 40 min |
| 14:00–14:40 | Vascular access | Visualisation of Superficial Veins and Epicutaneo-Caval Catheter (ECC) Insertion | 40 min |
| *14:40–15:10* | — | *Coffee break (video link check for the online lectures)* | *30 min* |
| 15:10–15:50 | Cardiac | The PDA: Echocardiographic Assessment of Haemodynamic Significance (live online) | 40 min |
| 15:50–16:30 | Cardiac | PDA Management: Pharmacological and Transcatheter Closure, with questions and close of Day 2 (live online) | 40 min |

# Day 3 — Wednesday 2 June 2027

## Vascular Access, GI, the Crashing Baby, Neuro, Asphyxia and Elective Workshops

| Time | Area | Topic | Duration |
|---|---|---|---|
| *08:00–08:15* | — | *Recap of Day 2* | *15 min* |
| 08:15–08:55 | Vascular access | Catheter Tip Localisation and Catheter-Related Complications: UVC, UAC and PICC; Malposition, Thrombosis and Effusions | 40 min |
| 08:55–09:40 | GI | Liver and Adrenals: Acute Abdominal Bleeds, incl. Adrenal Haemorrhage | 45 min |
| *09:40–10:00* | — | *Coffee break* | *20 min* |
| 10:00–10:40 | Vascular access | Prevention of CLABSI: Securement and Exit-Site Protection | 40 min |
| 10:40–11:20 | Emergency | The Crashing Baby Protocol | 40 min |
| 11:20–11:55 | Neuro | Neuro: Sinus Venous Thrombosis | 35 min |
| *11:55–12:00* | — | *Questions and discussion* | *5 min* |
| *12:00–13:00* | — | *Lunch break* | *60 min* |
| 13:00–13:30 | Cardiac | Haemodynamics in Asphyxia | 30 min |
| *13:30–13:40* | — | *Introduction to the elective workshops and participant allocation* | *10 min* |
| 13:40–14:30 | Elective | Elective Workshop — Part 1 | 50 min |
| *14:30–14:50* | — | *Coffee break* | *20 min* |
| 14:50–15:40 | Elective | Elective Workshop — Part 2 | 50 min |
| 15:40–16:30 | Elective | Elective Workshop — Part 3 | 50 min |
| *16:30–17:00* | — | *Workshop feedback, discussion and preparation for Day 4* | *30 min* |

## Elective workshops

Each participant stays in one workshop for all three parts. The workshops are informal, built around small-group discussion and clinical cases. Depending on the topic, they may include simple scanning exercises on fellow participants and EchoPAC measurement training.

- Cardiac — neonatal echocardiography and haemodynamics
- Lung — lung ultrasound
- GI/Abdominal — abdominal POCUS
- Vascular access — ultrasound-guided vascular access

# Day 4 — Thursday 3 June 2027

## Hands-On Workshops

08:00–16:15. Hands-on scanning and small-group discussions across all four topics, with all 60 participants working at the same time in 10 groups of 6.

Training takes place on:

- healthy children and infants (volunteers)
- simulators
- stable babies in the NICU, depending on availability on the day

## Faculty

- Prof. Yogen Singh, Los Angeles, USA — haemodynamics
- Prof. Alan Daneman, SickKids, Toronto, Canada — GI and NEC ultrasound
- Fiammetta Piersigilli, Brussels, Belgium — vascular access
- Nadya Yousef, Paris, France — lung ultrasound
- John Madar, Plymouth, UK — haemodynamics
- Instructors from Gothenburg, Stockholm, Halmstad and Borås